IEEE BIP 2026 · GUÁPILES, COSTA RICA · 11–13 NOVEMBER

BeeQ: Evaluating Quantum Fidelity Kernels as a New Approach Methodology for Molecular Toxicity Prediction

MATCHED COMPARISON | 893 MOLECULES | 10 SIMULATED QUBITS

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AGENDA

01 **Context**

Why honey-bee toxicity matters, and the research question that drives the work.

02 **Methodology**

The dataset, how molecular families are separated, and how each molecule is represented.

03 **Implementation**

What a kernel is, how descriptors are encoded into quantum states, and the scope of the simulation.

04 **Results**

Performance across model families, transfer beyond Development, and the external CR8 panel.

05 **Conclusions**

What the study does and does not show, and where it goes next.

THE TEAM

It started at Quantathon 2026 CR





There are far more substances than feasible assays

01 · EXPOSURE

Apis mellifera is a key pollinator

Agricultural active ingredients can reach it, and the acute lethal dose is an established regulatory criterion.

02 · MEASUREMENT

Measuring it requires animal testing

The in vivo assay gives the LD_{50} per bee, but it is slow, costly, and does not scale to the number of registered substances.

03 · PRIORITISATION

Not every molecule: the ones that matter

Models trained on already measured data flag which molecules are worth taking to the laboratory and which are not.

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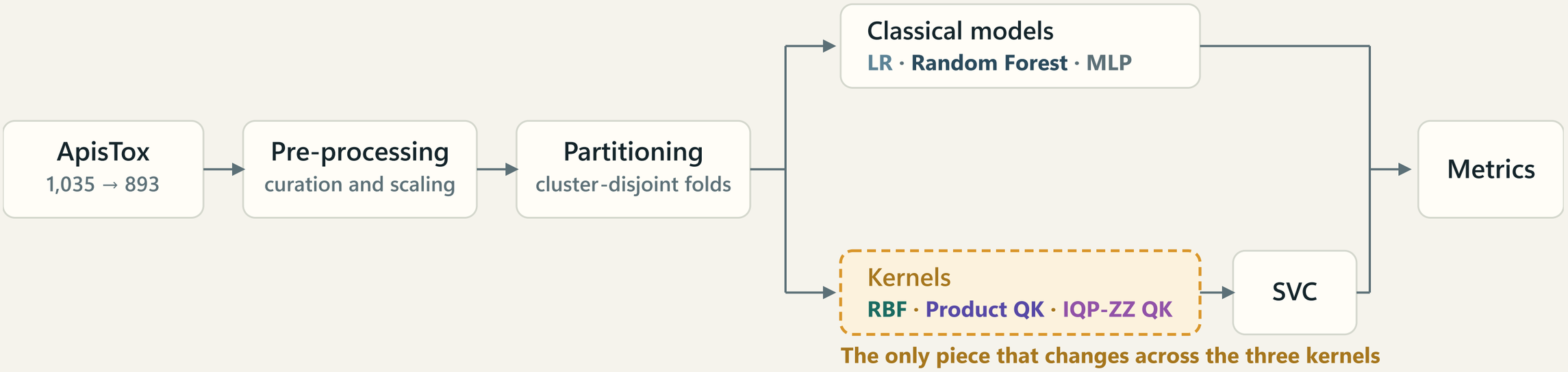
Models trained on already measured data flag which molecules are worth taking to the laboratory and which are not.

BeeQ sits at the third step: an exploratory prioritisation tool, not a regulatory decision criterion.



RESEARCH QUESTION

Does a fixed quantum fidelity kernel induce a predictively and geometrically different similarity structure when the molecular representation, partitions, pre-processing and evaluation are kept unchanged?



--- Conceptual schematic

893 curated molecules with measured acute toxicity

- **From 1,035 to 893**

  ApisTox contributes 1,035 compounds; additional curation and standardisation with RDKit leaves 893 organic molecules.

- **Binary classification, not regression**

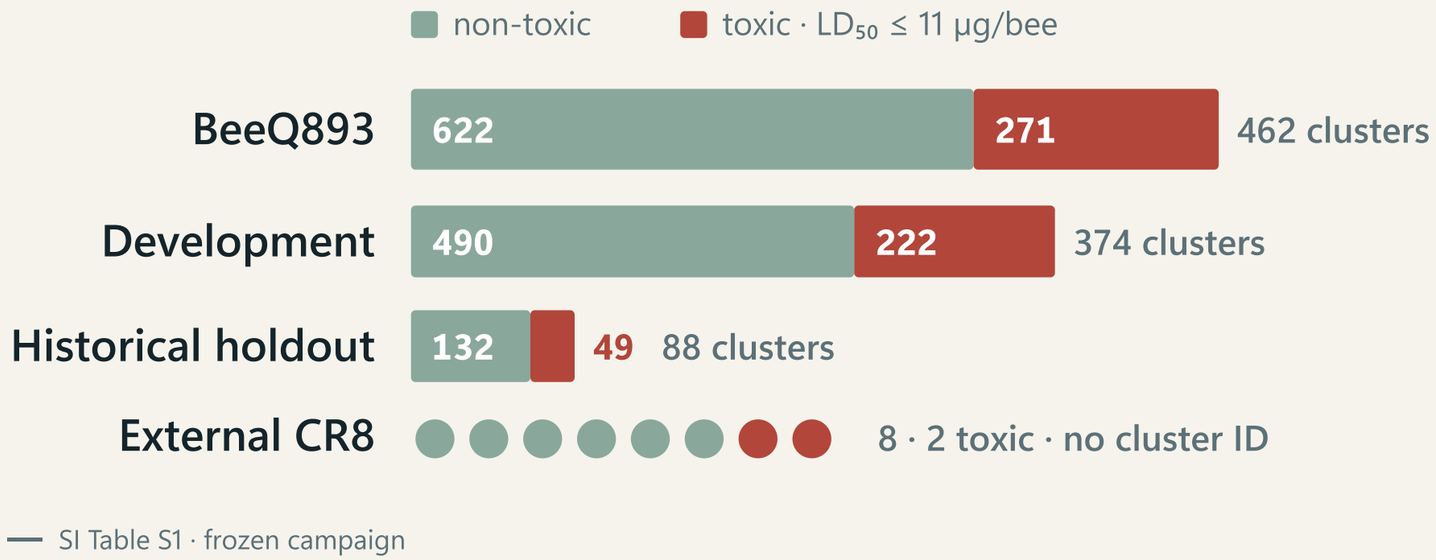
  The threshold was fixed before modelling and never tuned on performance.

THE ENDPOINT

$$y = 1 \text{ if } LD_{50} \leq 11 \text{ µg/bee}$$

$$y = 0 \text{ if } LD_{50} > 11 \text{ µg/bee}$$

893 curated molecules with measured acute toxicity



SI Table S1 · frozen campaign

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Nothing that is evaluated takes part in its own selection

1 · SELECTION

Settings are chosen using the training data only

i1 i2 i3 i4

never the part being evaluated

4 inner splits, grouped by cluster
hyperparameters by mean inner AUROC
threshold by maximum MCC

2 · FREEZE

One final refit, and from then on nothing changes again

all 712 Development molecules + frozen final threshold

manifest written beforehand

3 · EVALUATION

The external sets are opened exactly once

Historical holdout · 181

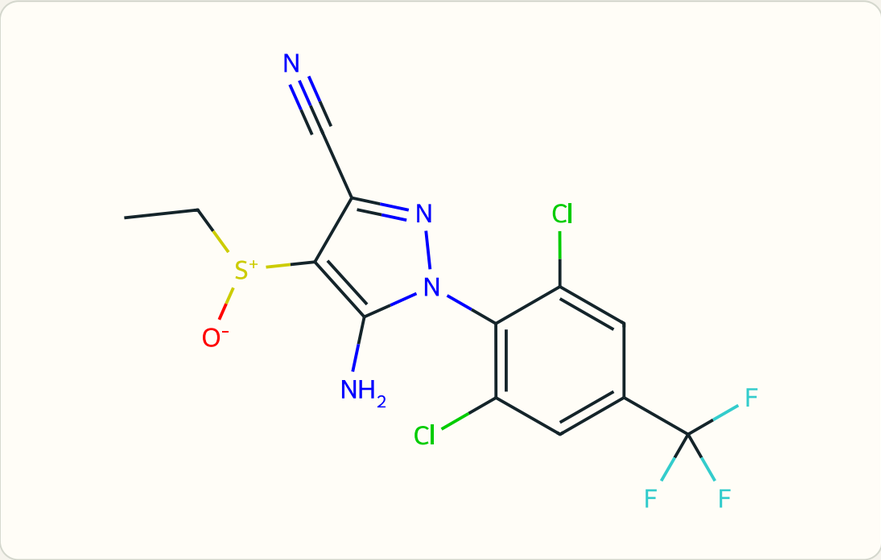
External CR8 · 8

no retuning, no second look

Whole clusters are never split between training and validation — a molecule's close relatives always stay on the same side.

— Morgan radius 2 · Tanimoto · Butina 0.60 → 462 clusters, 334 singletons, largest 40

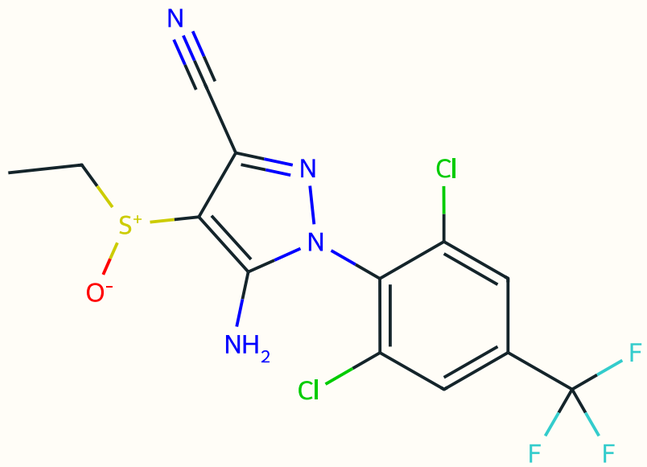
From a chemical structure to ten numbers



Ethiprole

— Structure drawn with RDKit from the frozen CR8 SMILES

From a chemical structure to ten numbers



Ethiprole

— Structure drawn with RDKit from the frozen CR8 SMILES

- **value** of the descriptor computed on the structure

z **standardised:** (value − mean) / SD, with Development mean and SD

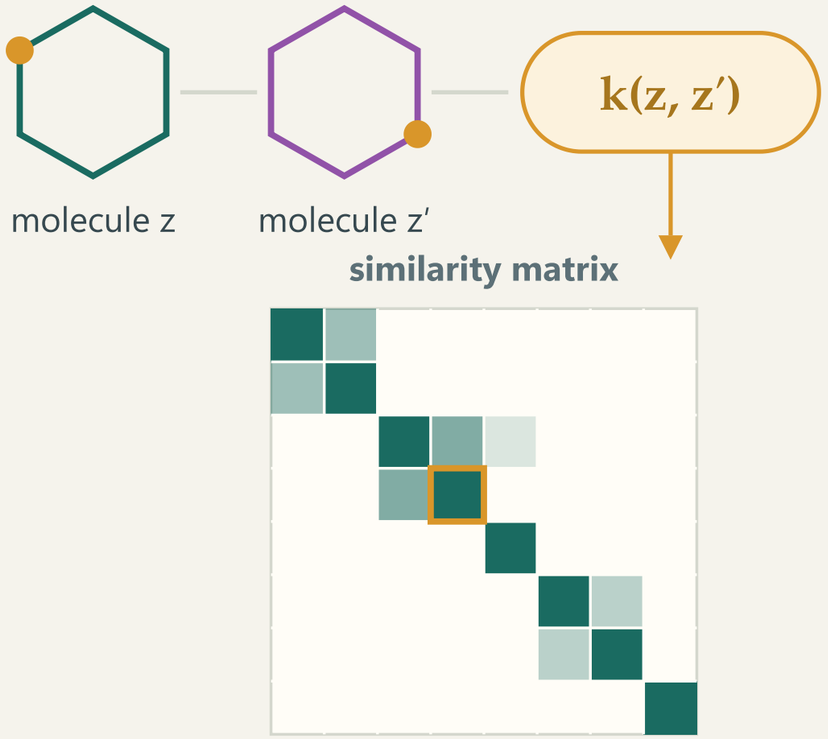
θ **angle:** θ = s · z, what rotates the qubit, with s = 0.25

A kernel is a way of measuring likeness

Instead of asking the model to reason about each molecule on its own, we hand it a table with the likeness of **every pair** of molecules.

$$K_{RBF}(z, z') = \exp(-\gamma \, \|z - z'\|^2)$$

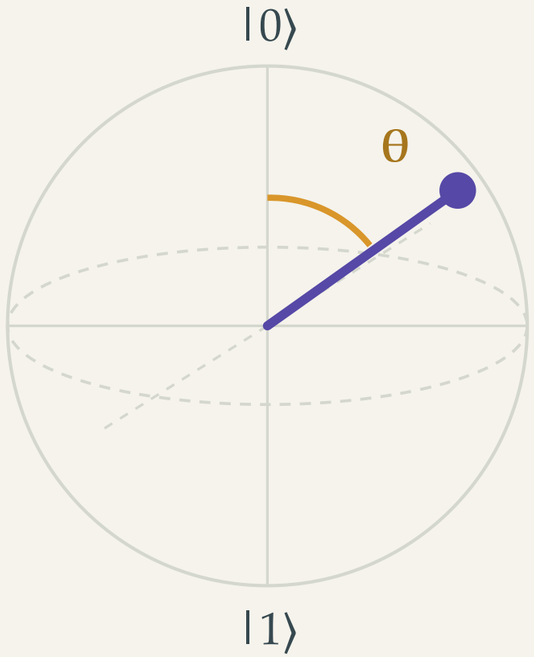
The classical reference. It decays with the distance between standardised descriptors; γ controls how fast.



One pair of molecules fills one cell.

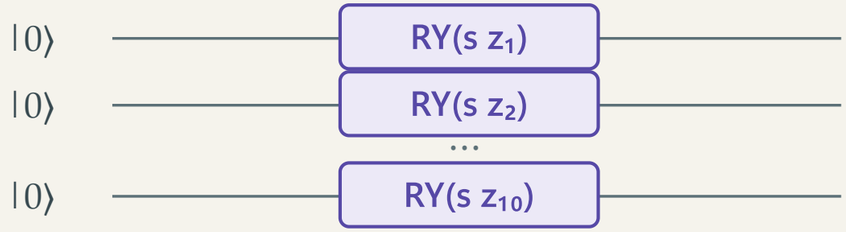
Each descriptor rotates a qubit; likeness is the fidelity

LIKENESS = FIDELITY $K_Q(z, z') = |\langle\psi(z)|\psi(z')\rangle|^2$ Between 0 and 1; it is 1 when the two states coincide.



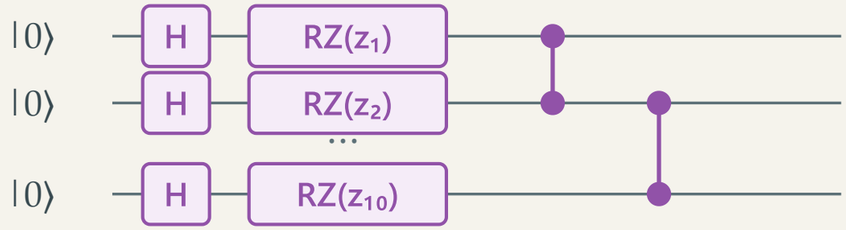
$\theta_i = s \cdot z_i \cdot$ one qubit per descriptor

PRODUCT · s = 0.25



no entanglement

IQP-ZZ · s = 0.125 · λ = 1



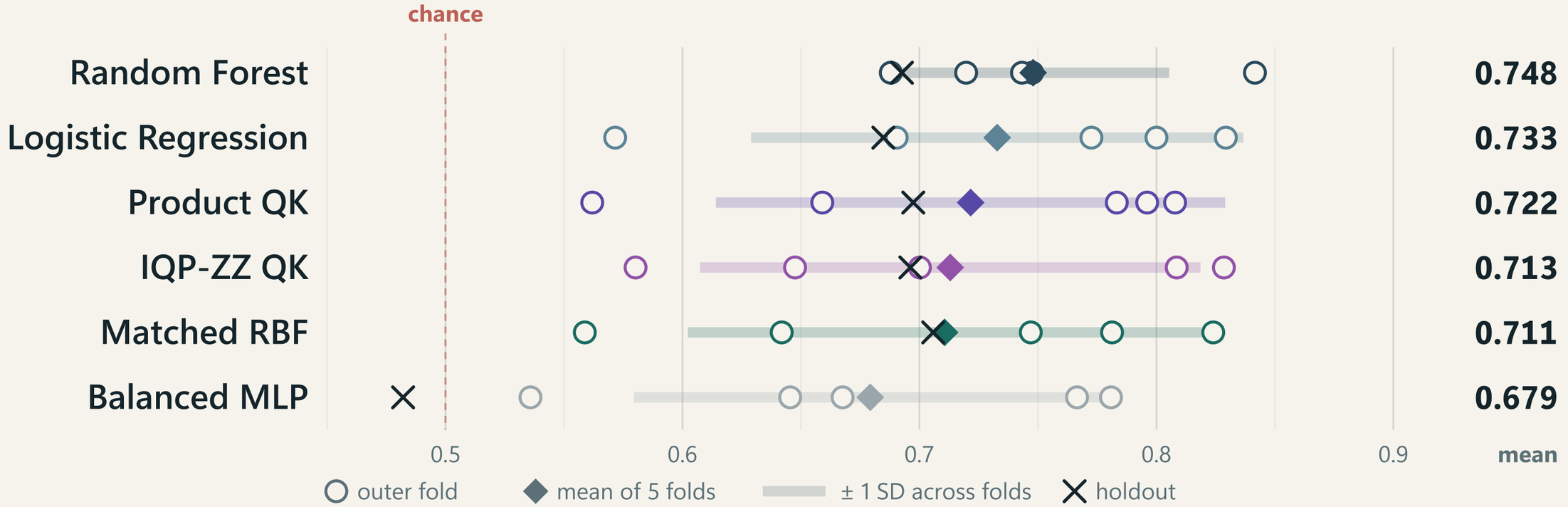
RZZ chain
nearest neighbours
no final Hadamard

--- Conceptual schematics

- Exact statevectors in NumPy
- 1,024 amplitudes per molecule
- No shots, no noise, no quantum hardware

Dispersion across folds exceeds the differences between models

Dispersion across folds exceeds the differences between models



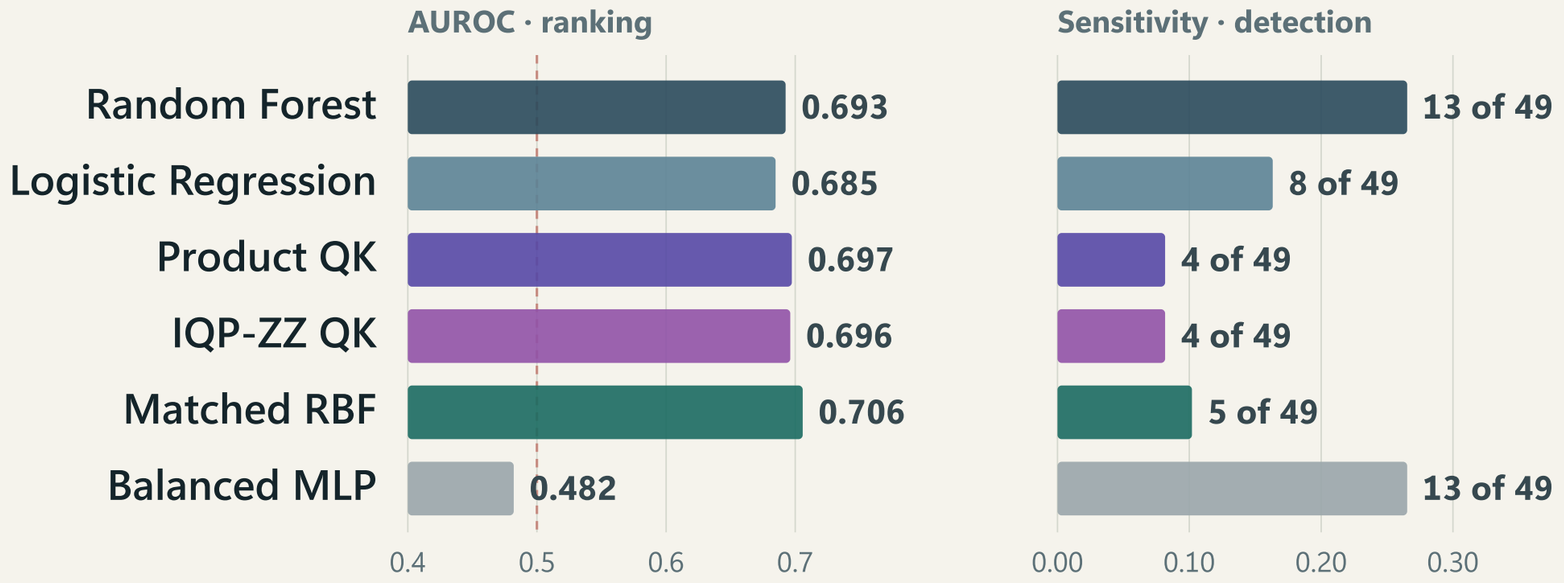
Random Forest reached the highest mean: 0.748

AUROC — ranking, not accuracy

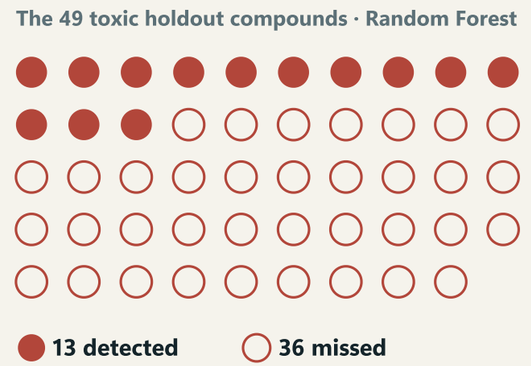
The probability that a random toxic molecule scores above a random non-toxic one.

— Paper Table I · SI Tables S17 and S18

Ranking well and detecting little are compatible

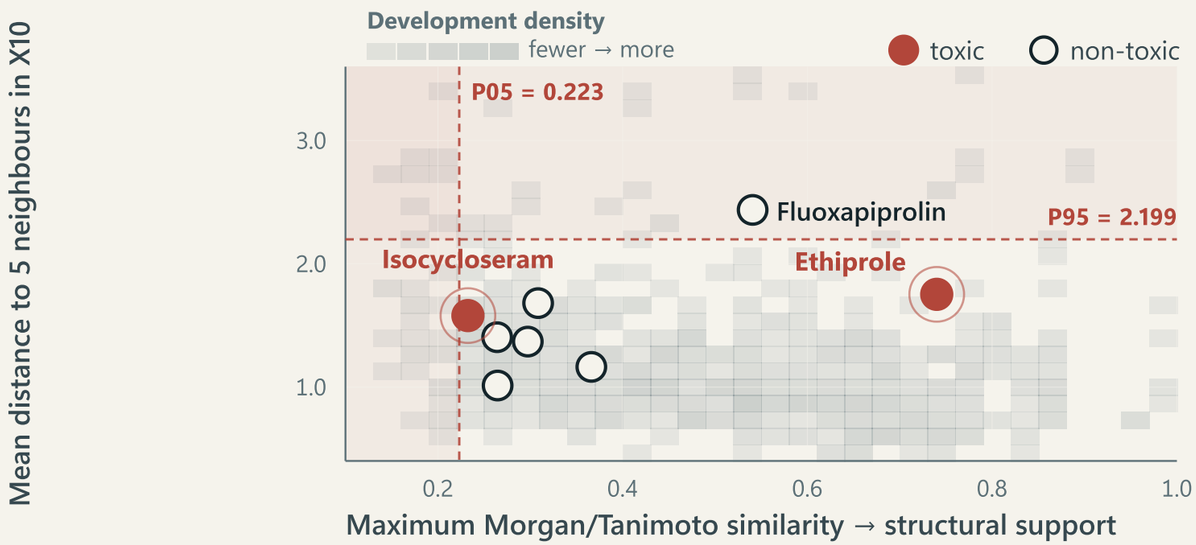


— SI Table S18



Five models hold AUROC near 0.70 and still let most of the toxic compounds through.

Being inside the domain does not guarantee a correct call



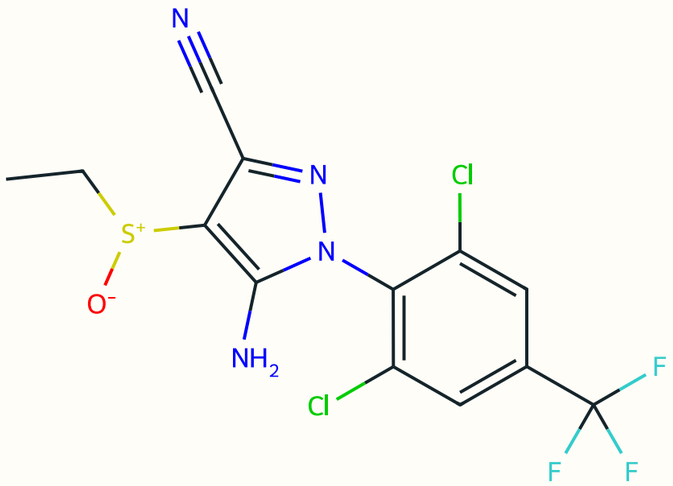
SI Table S26

	RF	LR	Prod	IQP	RBF	MLP
Isotianil	○	○	○	○	○	○
Fluoxapiprolin	○	○	○	○	○	✕
Ethiprole	✕	✕	✕	✕	✕	✕
Cyclobutrifluram	○	○	○	○	○	○
Isoprothiolane	○	○	○	○	○	○
Isocycloseram	✕	✕	✕	✕	✕	○
Spiropidion	○	○	○	○	○	○
Florylpicoxamid	○	○	○	○	○	✕

○ correct decision ✕ error frozen thresholds, no retuning

SI Table S27 · balanced MLP

A nearly identical toxic neighbour was not enough



Ethiprole

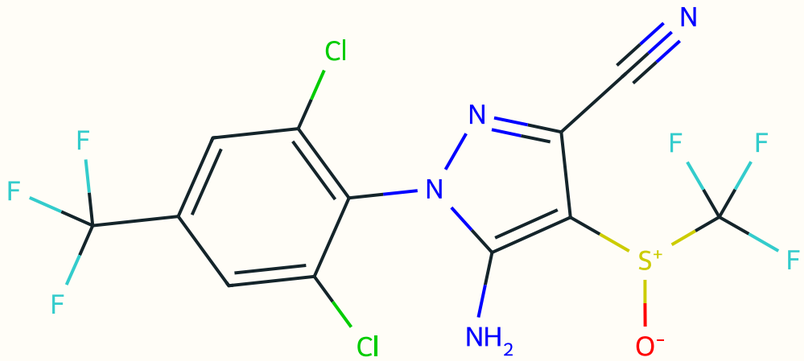
CR8 · external

● toxic

predicted non-toxic ×6

0.740

Morgan/Tanimoto similarity



Fipronil

Development · ID 387

● toxic

in training

Two findings, and a clear boundary

WHAT WE FOUND

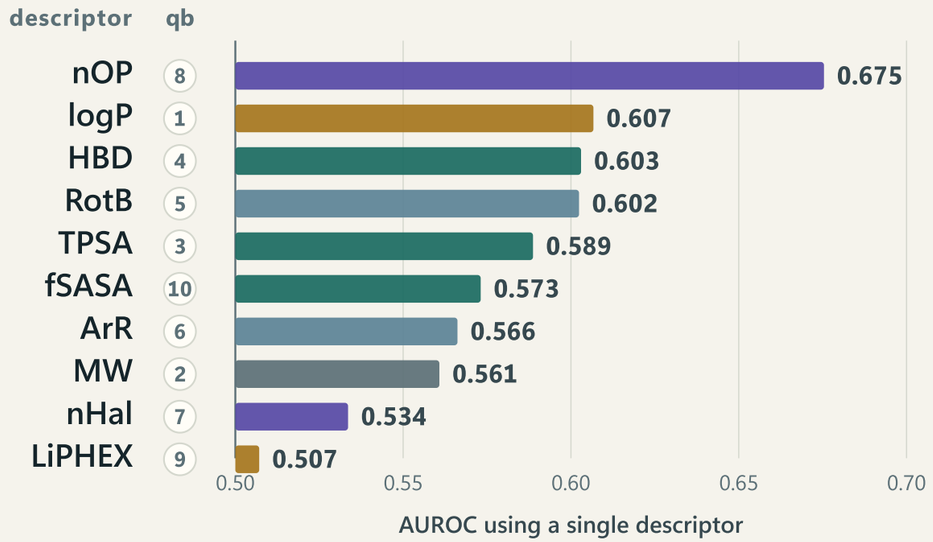
- **No consistent predictive improvement**
  Both fidelity kernels landed very close to the matched RBF.
- **No clearly distinct geometry**
  The Gram matrices turned out highly aligned with each other.
- **Detection remains the challenge**
  Beyond Development, sensitivity falls to 0.082–0.265.

WHAT THIS WORK DOES NOT SHOW

- It does not demonstrate quantum advantage
- It does not evaluate noise or real quantum hardware
- It does not establish clinical or regulatory utility

Separating kernel changes from representation changes

- **Optimise the encoding scale**
  Bandwidth turned out to be the decisive factor.
- **Train the embedding by kernel-target alignment**
  Instead of fixed maps, learn the map's parameters.
- **Rearrange and widen the qubits**
  The RZZ chain couples neighbours only: which descriptor sits next to which matters.



SI Table S5 · descriptive

Thank you.
Questions?

DATA AND CODE

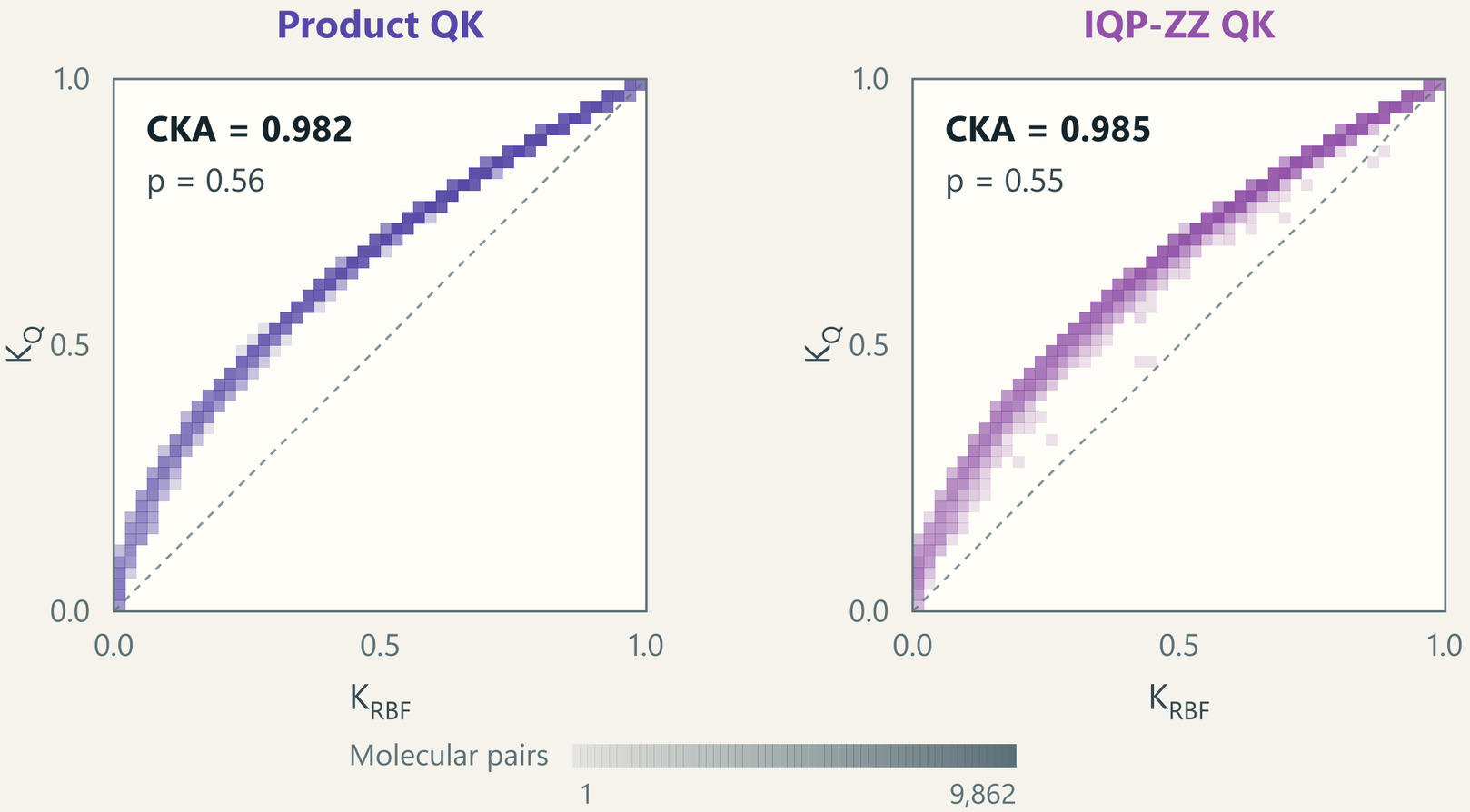
Supporting Information on Zenodo
doi.org/10.5281/zenodo.23329579

Dataset, descriptors and training workflow
github.com/andresquesadag/BeeQ

Scan for the repository

SUPPLEMENTARY KERNEL GEOMETRY

The two quantum kernels against the matched RBF



All 253,116 unique Development pairs

SUPPLEMENTARY X10 IN DETAIL

What each of the ten descriptors is

QUBIT	SYMBOL	COLUMN	DEFINITION	UNIT
1	**logP**	MolLogP	Lipophilicity: octanol/water partition estimated by the Wildman–Crippen method.	—
2	**MW**	MolWt	Average molecular weight.	g/mol
3	**TPSA**	TPSA_SP	Topological polar surface area, including sulfur and phosphorus contributions.	$Å^2$
4	**HBD**	NumHDonors	Hydrogen-bond donor count.	count
5	**RotB**	NumRotatableBonds	Rotatable-bond count: a proxy for molecular flexibility.	count
6	**ArR**	NumAromaticRings	Aromatic-ring count.	count
7	**nHal**	nHalogen	Number of F, Cl, Br and I atoms.	count
8	**nOP**	n_OP	Matches to an organophosphorus-like SMARTS motif. It is **not** the phosphorus count: several matches can involve the same atom.	count
9	**LiPHEX**	LiPHEX_prediction	n-hexadecane/water partition predicted by the in-house LiPHEX model: a more apolar lipophilicity regime than logP.	—
10	**fSASA**	sasa002_frac_polar_hetero_only	Fraction of the solvent-accessible surface assigned to polar heteroatoms (O, N, S, P): a 3-D measure of polar exposure.	fraction

The descriptor order fixes the qubit assignment